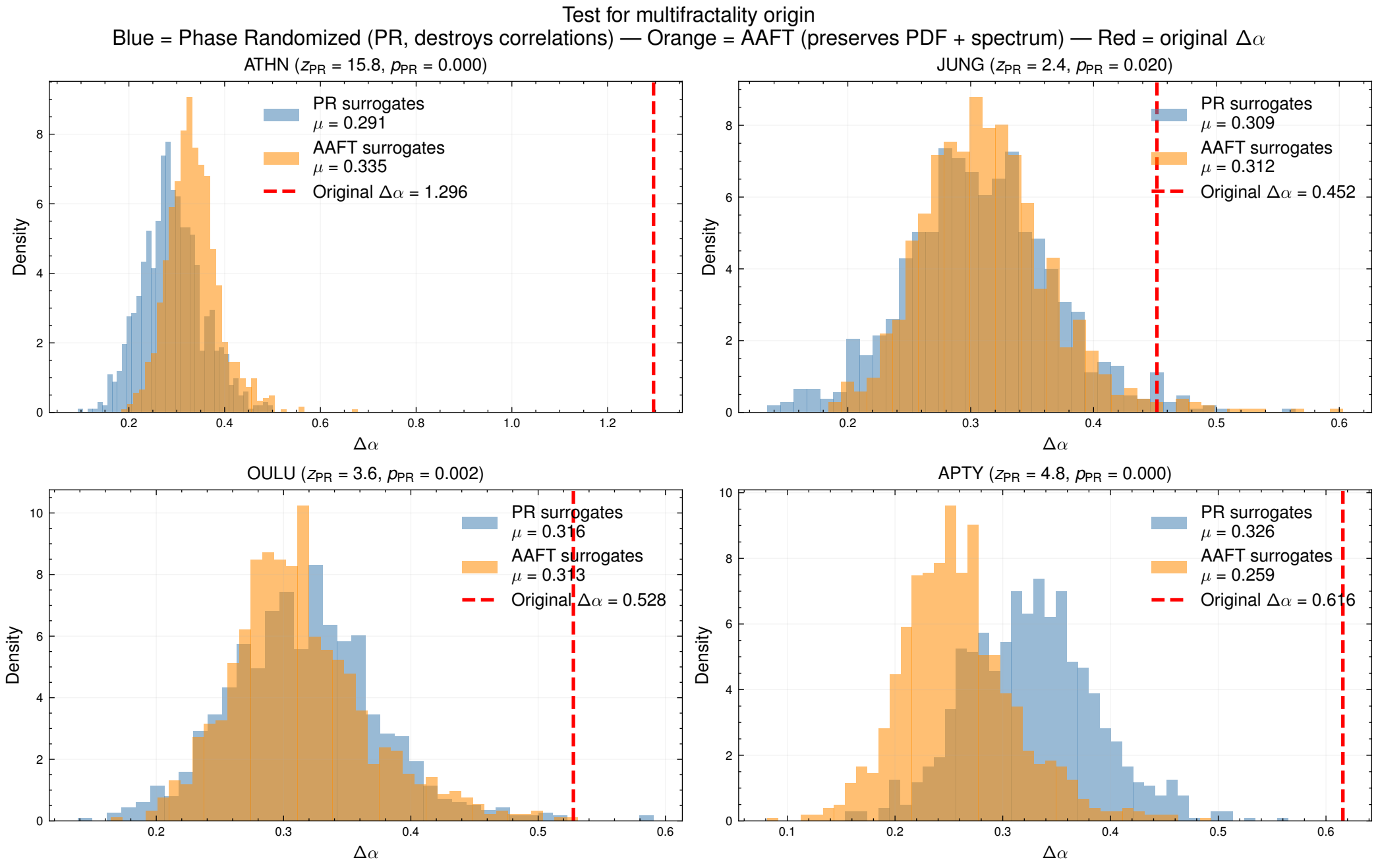
Test for multifractality origin
Blue = Phase Randomized (PR, destroys correlations) — Orange = AAFT (preserves PDF + spectrum) — Red = original $\Delta\alpha$
ATHN (z_{PR} = 15.8, p_{PR} = 0.000)
PR surrogates
μ = 0.291
AAFT surrogates
μ = 0.335
Original $\Delta\alpha$ = 1.296
JUNG (z_{PR} = 2.4, p_{PR} = 0.020)
PR surrogates
μ = 0.309
AAFT surrogates
μ = 0.312
Original $\Delta\alpha$ = 0.452
OULU (z_{PR} = 3.6, p_{PR} = 0.002)
PR surrogates
μ = 0.316
AAFT surrogates
μ = 0.313
Original $\Delta\alpha$ = 0.528
APTY (z_{PR} = 4.8, p_{PR} = 0.000)
PR surrogates
μ = 0.326
AAFT surrogates
μ = 0.259
Original $\Delta\alpha$ = 0.616
Density
$\Delta\alpha$